

B2 R2 Responses Master

Paper: Governance Concentration Beyond Token Allocation: A Protocol-Address-Corrected Cross-Sectional Audit of 52 DePIN and DeFi Protocols **Working title (5 words):** Governance Concentration Beyond Token Allocation **Author:** Zach Zukowski (Tokenization Systems; zach@tokenization.systems) **Journal:** Frontiers in Blockchain (Manuscript ID 1853465; SSRN 6599278) **Round:** R2 (response to Reviewer 1 Round 2 feedback) **Date:** 2026-05-31

Preamble

We thank both reviewers for their continued engagement with the manuscript. Reviewer 2 endorsed publication in R2. The responses below address Reviewer 1's four Round 2 issues at the level of granularity raised. Reviewer 1's general framing – “reproduce all the tables and figures, and make the text reflect the current data” – was adopted as a discipline for the entire revision, which extends beyond the four flagged issues into a comprehensive reproduce-and-sync pass and a structural reframe of the manuscript around its empirical contributions.

The most consequential single result of the reproduce-and-sync is a substantive interpretive change in Section 4.5: with corrected post-exclusion holding HHI baselines applied consistently across the holding-concentration table and the delegation table, Uniswap and Optimism flip from delegation-mediated dispersion to delegation amplification. The revised paper documents a predominant-amplification finding with structural exceptions: of the eighteen protocols with computable voting concentration, thirteen amplify and five disperse. The dispersers are ENS (Section 4.5.1, mature delegate program), GMX (Section 4.5.1.1, methodology-sensitive counterexample), Helium (Section 4.5.4, Solana VSR-lockup-weighted), Jupiter (Section 4.5.4, Solana delegated-vote with broad-airdrop user base; the most-extreme dispersion outlier), and Livepeer (Section 4.5, the most pronounced DePIN governance disperser).

A PCA-symmetric robustness check (Section 4.6) extends the predominant-amplification finding across three independent governance-data sources: Tally all-delegates depth (AAVE N = 150,911; COMP N = 18,627; UNI N = 48,707; ARB N = 437,453; all four sub-0.005 HHI shift versus the top-1000 baseline, validating the top-1000 methodology floor); Snapshot signer-side functional-PCA exclusion registering four confirmed entries (Compound Foundation; Balancer DAO multisig; Kevin Owocki / Bitcoin Maintainer; WXM Hedgey-vested insider); and Solana on-chain governance program parsing for Jupiter, Helium, and Drift. A methodologically substantive finding from the Snapshot signer-side PCA-symmetric recompute: exclusion is not monotonic in the concentration-reduction direction. For three of four Snapshot protocols with confirmed signer-side functional PCAs, exclusion produces the expected concentration reduction; for GTC, signer-side exclusion increases voting HHI from 0.1792 to 0.1948 (smaller-denominator effect). PCA-symmetric exclusion is therefore a substantive direction-of-effect test rather than a mechanical concentration-reduction adjustment.

The manuscript was also reframed around its empirical contributions. The R1 framing developed a five-lens philosophical framework integrated with the empirical results; the revised manuscript leads with the measurement contributions and routes the philosophical-interpretation thread to a

companion paper (Zukowski 2026e). This preserves every R1 statistical finding while making the empirical contribution the spine of the paper.

Issue 1: Table 7 holding-HHI inconsistency (interpretive flip)

Reviewer comment (verbatim, key points)

Table 7 still uses the old holding-HHI baselines for some protocols, even though the corresponding HHI values have been revised elsewhere in the paper and repository. ... Using the revised holding-HHI value would make the voting/holding ratio roughly 2.7X rather than 0.84X [for Uniswap]. Optimism's Table 4/processed-data holding HHI is approximately 0.009, while Table 7 still uses 0.042. ... Lido's ratio also appears lower when the revised holding HHI is used. ... I recommend that Table 7 be fully recomputed using the same proper holding-HHI values used in Table 4 and in the regression dataset.

Response

The reviewer is correct on all three flagged cases. The delegation table has been fully recomputed using post-exclusion holding HHIs consistent with the current holding-concentration table and regression dataset. The Compound and Arbitrum rows use Snapshot-sourced voting HHIs per the R1 author rationale (larger active-voter pools than Tally top-100 sampling); the remaining voting-HHI source assignments reflect the methodology floor of all-proposal Snapshot pulls and top-1000-delegate Tally pulls. Holding HHIs are uniformly post-exclusion across the table.

The revised delegation table (Table 6) reports eighteen protocols. Relative to R1 it adds the protocols for which on-chain voting concentration is computable on a proper-weight basis, including three Solana-native protocols (Jupiter via per-signer max-weight aggregation on delegated votes; Helium and Drift via SPL Governance VoteRecordV2 parsing), and it corrects the holding-HHI baselines for the carried-over rows. Of the eighteen, thirteen amplify (voting HHI greater than holding HHI) and five disperse. The amplification range runs from 1.6x (Drift) to 25.6x (Polkadot), with mean approximately 6.8x and median 4.4x. The thirteen-protocol core of the amplifying group includes DIMO, Lido, WeatherXM, Compound, Aave, Uniswap, Arbitrum, Optimism, and Drift; the five dispersers are ENS (0.45x), GMX (0.87x), Helium (0.35x to 0.53x), Jupiter (0.043x, the most-extreme outlier), and Livepeer (0.27x).

Selected post-exclusion rows (illustrative; full table in the manuscript):

Protocol	Sector	Holding HHI (post-exclusion)	Voting HHI	Ratio	Direction
DIMO	DePIN	0.025	0.228	9.1x	amplifies
Lido	DeFi	0.008	0.050	6.3x	amplifies
WeatherXM	DePIN	0.148	0.556	3.8x	amplifies
Compound	DeFi	0.009	0.089	9.9x	amplifies

Protocol	Sector	Holding HHI (post-exclusion)	Voting HHI	Ratio	Direction
Aave	DeFi	0.013	0.058	4.4x	amplifies
Uniswap	DeFi	0.010	0.027	2.7x	amplifies (FLIP from R1 0.84x)
Arbitrum	Infra	0.012	0.038	3.1x	amplifies
Optimism	Infra	0.009	0.033	3.6x	amplifies (FLIP from R1 0.79x)
GMX	DeFi	0.065	0.057	0.87x	DISPERSE S (Section 4.5.1.1; methodolo gy- sensitive)
ENS	Infra	0.049	0.022	0.45x	disperses (Section 4.5.1; mature delegate program)
Drift	DeFi	0.053	0.083	1.6x	amplifies (Solana VSR; Section 4.5.4)
Helium	DePIN	0.075	0.026 to 0.039	0.35x to 0.53x	DISPERSE S (Solana VSR; Section 4.5.4)
Jupiter	DeFi	0.126	0.0055	0.043x	DISPERSE S (Solana delegated- vote; most- extreme outlier; Section 4.5.4)

Solana rows use on-chain governance program parsing (Jupiter via Dune `jupiter_solana.govern_call_setvote` per-signer max-weight aggregation; Helium and Drift via Helius RPC parsing of SPL Governance VoteRecordV2 accounts). The Aave row

reflects an all-delegates pull at $N = 150,911$; the four deeper- N sweeps validate the top-1000 methodology floor (all four sub-0.005 HHI shift versus the top-1000 baseline).

Substantive interpretive consequence. The flip is not a typo correction: it changes the Section 4.5 finding from the R1 “delegation amplifies unevenly, with two infrastructure protocols showing opposite patterns” to the revised predominant-amplification framing, in which thirteen of eighteen protocols amplify, with five structural exceptions documenting design-tractable dispersion. The Section 4.5 opening paragraph, sector-breakdown paragraph, Citizens’ House and Dual Governance paragraphs, and the Fritsch et al. extension paragraph have all been rewritten. New subsections Section 4.5.1 (ENS counterexample), Section 4.5.1.1 (GMX counterexample), Section 4.5.4 (within-Solana governance heterogeneity), Section 4.5.4.1 (Drift Foundation proposal-authorship versus vote-weight decoupling), and Section 4.5.5 (cross-protocol institutional-governance-investor pattern as candidate fifth concentration axis) have been added. The abstract and the Section 6 conclusion are updated to the predominant-amplification framing.

On Lido. The reviewer’s observation that Lido’s ratio “appears lower” with the revised holding HHI is opposite to what our data produces. Applying PCA exclusion of the full set of Lido DAO addresses yields a Lido holding HHI of 0.008. The Snapshot voting pull, taken across all proposals in the analysis window ($N = 333$), yields voting HHI 0.050 and an amplification ratio of 6.3x. If the reviewer is computing a different Lido holding HHI from an alternative exclusion list (for example, one that does not exclude the Lido DAO Aragon treasury), we welcome specification of the methodology; the exclusion set is documented in Supplementary File S6 at address granularity.

On the Uniswap counterexample. The pre-correction Uniswap result (0.84x) had been cited as evidence that Uniswap’s mature delegate program disperses governance power. With the burn-rule exclusion applied (the 0x000...000dead address holding 102.5M UNI, 11.3 percent of supply, structurally unable to participate in governance under any condition), Uniswap joins the predominant-amplification pattern at 2.7x. We document this transition explicitly in Section 4.5 because the prior result was load-bearing in the dispersion-via-mature-delegate-program literature; the corrected analysis reflects the cost of including a wallet that structurally cannot vote in the holding denominator.

Compound Foundation as PCA at the voting layer. Fact-checking of the Tally API surfaced a substantive methodological finding: the top COMP Tally delegate (21.5 percent of total delegated voting power in the May 2026 pull) is Compound Foundation itself, structurally a protocol-controlled entity. The PCA-symmetric voting-HHI robustness check (Section 4.6) extends the holding-side methodology to the voting layer: PCA exclusion is not monotonic in the concentration-reduction direction (see the GTC counterexample above), but predominant amplification holds for the deeper- N Tally sweep across AAVE, COMP, UNI, and ARB at sub-0.005 HHI shift versus the top-1000 baseline.

Manuscript changes: Delegation table rows fully recomputed; table caption updated with explicit post-exclusion methodology note; Section 4.5 opening through Fritsch-extension paragraphs rewritten; new Sections 4.5.1, 4.5.1.1, 4.5.4, 4.5.4.1, 4.5.5; Section 4.6 PCA-symmetric robustness paragraph added; Section 3.3 snapshot-date discipline paragraph added (documents the March 2026 to May 2026 delegate-pool drift; the revised manuscript uses the

March 2026 snapshot consistent with the rest of the dataset for the protocols with March 2026 voting data; the GMX and ENS additions use May 2026 Tally per the delegation-table footnote); Figure 4 (delegation-adjusted bar chart) regenerated with the eighteen-protocol data; abstract, Section 1.1, and Section 5.6 updated to the predominant-amplification framing.

Issue 2: Curve Top-10 column + CRV text disambiguation

Reviewer comment (verbatim)

In Table 4, Curve's HHI has been corrected to the post-exclusion raw-CRV value, but its Top-10 value is still the pre-exclusion one. The repository reports a substantially lower post-exclusion Top-10 value for Curve than the value shown in the table. In several places in the text, CRV is reported as HHI = 0.171, even though Table 4 now reports raw CRV, with HHI = 0.017. If this is supposed to be about veCRV, please mention it explicitly in the text.

Response

The reviewer is correct on both points. Table 4 Curve Top-10 percentage has been corrected from 60.7 percent (pre-Convex-exclusion) to the post-exclusion value matching the regression dataset. The CRV-versus-veCRV disambiguation has been implemented across all textual surfaces, with the 0.171 proxy replaced by a directly computed veCRV voting concentration value (0.26).

Direct veCRV voting concentration computation. The R1 manuscript used 0.171 as a proxy for veCRV-weighted voting concentration. The 0.171 figure is the raw CRV holding HHI including the Convex veCRV custody position (which holds substantial CRV that has been locked into veCRV). The revised paper computes the actual veCRV-weighted voting concentration directly via Dune analysis of the Curve VotingEscrow Deposit events at 0x5f3b5dfb7b28cdbc7faba78963ee202a494e2a2. Aggregating cumulative CRV deposits across the top-50 historical veCRV lockers yields HHI 0.26: Convex CRV Locker at 48.0 percent of top-50 deposit-weighted share; Yearn yveCRV at 13.5 percent; Stake DAO sdCRV at 9.5 percent; the remaining 47 lockers each less than 4 percent. The 0.26 direct computation captures the actual gauge-weight voting concentration that veCRV produces and is substantively higher than the 0.171 proxy.

Disambiguation applied at every textual surface.

- Table 3 row: HHI = 0.014 (raw post-Convex-exclusion CRV holdings); Top-10 percent = 31.2 (post-exclusion).
- Table 3 caption: HHI for Curve reflects raw CRV holdings after excluding the Convex veCRV custody position; veCRV-weighted voting concentration is approximately 0.26 computed from cumulative CRV deposit volume across the top-50 historical veCRV lockers; methodology discussed in Section 4.5.2.
- Section 4.2 cross-protocol range description: CRV's raw 0.014 places it mid-range; the 0.26 veCRV voting concentration is referenced with explicit methodology cite to Section 4.5.2.

- Section 4.5.2 (vote-escrowed governance subsection; new in R2): Both Curve veCRV (15x amplification relative to raw 0.014) and Balancer veBAL (21x amplification at voting HHI 0.626 against holding 0.030) documented as a distinct extreme-amplification mechanism class separate from delegate-program-based amplification.

The Aethir Table 3 row was also updated. The R1 footnote value reflected an earlier snapshot; the revised direct Dune pull yields HHI 0.095 post-PCA-exclusion (the 0xfc78 Safe multisig was identified as an additional PCA). Both reflect raw top-1000 holder concentration after PCA exclusion; the difference between the earlier and the direct pulls is documented in Supplementary File S6.

Manuscript changes: Table 3 CRV row corrected; Table 3 methodology notes updated with explicit veCRV decomposition (Convex 48 percent, Yearn 14 percent, Stake DAO 9 percent); new Section 4.5.2 “Secondary extension: vote-escrowed governance” documenting both Curve and Balancer ve-token amplification; Supplementary File S6 (address-by-address PCA documentation) updated with the Aethir 0xfc78 entry; a supplementary exhibit reporting the veCRV voting-concentration computation ships with the supplementary materials.

Issue 3: Table 5 N = 40 and TT-expanded subsidy reproducibility

Reviewer comment (verbatim)

First, the HHI-Gini row reports N = 40, but the current data seem to contain missing Gini values for some protocols. Second, the TT-expanded subsidy analysis is still not straightforwardly reproducible from the repository files. The CSV contains fewer observations than the N reported in the manuscript. In addition, the conclusions drawn appear to be based on the old numbers.

Response

The reviewer is correct on both sub-issues. Both have been addressed.

HHI-Gini row N = 40. Three DePIN protocols (Aethir, Hivemapper, io.net) appeared in the R1 Table 4 footnote with HHI values but were noted as lacking the holder-level Gini and percentile data required for Table 4 reporting. The R1 manuscript inconsistently reported N = 40 for the HHI-Gini correlation despite this gap. The revised paper resolves the gap by pulling full holder-level data for all three:

- Hivemapper (HONEY; Solana SPL): pulled via Helius DAS API (90,680 unique holders). HHI 0.018, Gini 0.91, Top-1 percent 5.7, Top-10 percent 31.1. Added as a full Table 3 row.
- io.net (IO; Solana SPL): pulled via Helius DAS API (84,861 unique holders). HHI 0.125, Gini 0.94, Top-1 percent 33.5, Top-10 percent 61.2. Added as a full Table 3 row.
- Aethir (ATH; Ethereum ERC-20): pulled via Dune top-1008 with eight PCA exclusions. HHI 0.095, Gini 0.94, Top-1 percent 19.7, Top-10 percent 67.6. Added as a full Table 3 row.

All three protocols now appear as full cross-section rows (Table 3) with complete holder-level data. The revised manuscript reports the HHI-Gini correlation on the $N = 48$ post-exclusion comparable cohort, the frame for which holder-level Gini is directly comparable (Pearson $r = 0.52$, $p < 0.001$; Table 4 and Supplementary File S0); the comparable cohort grew from 40 to 48 as the cross-section reached its final 52-protocol frame.

TT-expanded subsidy CSV reproducibility. The reviewer correctly identifies that the R1 manuscript's reported N for the TT-expanded subsidy test does not match the row count of subsidy-ratio-populated observations in the underlying CSV. In the revised paper the subsidy battery is reported on a single regression dataset: the dataset contains 20 protocols with non-null Token Terminal subsidy data, while the headline subsidy battery uses $N = 23$ protocols with non-zero subsidy under either Token Terminal or on-chain metrics (and $N = 22$ when Livepeer is excluded). The revised values are:

- Table 5 "Subsidy ratio (cross-sector)" row: Pearson $r = 0.62$, $p = 0.002$, $N = 23$. Fragile (Livepeer-driven).
- Table 5 "Subsidy ratio (TT expanded)" row: Pearson $r = 0.12$, $p = 0.61$, $N = 20$. Null (robustness check).
- Table 5 "Subsidy ratio (excluding Livepeer)" row: Pearson $r = 0.07$, $p = 0.76$, $N = 22$. Null.
- Spearman rank correlation on the full subsidy sample: $\rho = 0.26$, $p = 0.23$.
- Section 4.6 Robustness paragraph: the headline subsidy correlation is entirely Livepeer-driven (Livepeer subsidy ratio of 88.5x is a 3.5-sigma outlier on the subsidy axis); excluding Livepeer alone reduces the Pearson correlation to a clean null ($r = 0.07$).

A new multivariate specification (Section 4.6) with both subsidy ratio and DePIN sector indicator on the 23-protocol non-zero-subsidy sample (HC3 robust standard errors) reinforces the Livepeer-driven interpretation: with all 23 protocols, the DePIN sector indicator is significant ($p = 0.016$) while subsidy is not ($p = 0.36$; Adj R-squared = 0.47); excluding Livepeer, the subsidy coefficient is strongly non-significant ($p = 0.94$) while the DePIN sector indicator is significant ($p = 0.007$). A convention-invariance robustness check using the on-chain operating-cost subsidy specification reproduces the same null pattern (subsidy $p = 0.45$; DePIN $p = 0.005$).

On-chain operating-cost subsidy methodology. A focused investigation against the raw on-chain operating-cost computations for the burn-active panel surfaced four methodology refinements grounded in project-Foundation primary sources (DIMO, Morpheus AI, Filecoin, Helium). Material reclassifications: DIMO moves from the net-deflationary subset to the subsidy-heavy subset (an earlier burn-to-dead aggregation diverged from the DIMO Foundation's revenue methodology by 13.8x token count); Morpheus AI moves from net-deflationary to subsidy-heavy (an earlier filter captured cross-chain bridge flows rather than protocol revenue and emissions); Filecoin subsidy ratio refined to 46.05 (Token Terminal emissions undercount of 140 percent due to a vesting-discount filter; cross-validated against Spacescope, Tokenomist, and the Coinbase Institutional Tokenomics Review primary sources within 1.8 percent). A verification pass against the remaining thirteen panel protocols produced zero additional methodology shifts. The cross-protocol pattern is structurally informative: the four protocols requiring methodology refinement are all DePIN-class; the eleven non-DePIN protocols (DeFi blue chips, mid-cap DeFi, L2 / L1 infrastructure) passed direct verification with

no methodology shifts. This is offered as a candidate methodology contribution; the structural difference between newer DePIN protocols (cross-chain bridging, custom reward distribution mechanisms) and mature non-DePIN protocols (community-aggregator alignment, methodologically stable Foundation reporting) may motivate further work in subsequent B-series papers.

Manuscript changes: Table 5 rows updated to current values; the cross-section (Table 3) reaches its final $N = 52$ frame, with Hivemapper, io.net, and Aethir among the full rows; Section 4.6 subsidy-multivariate paragraph documenting the four specifications; Supplementary File S6 documents the on-chain operating-cost methodology per protocol; Supplementary File S8 documents the Token Terminal expansion battery.

Issue 4: Figure-vs-text statistical drift (Figures 2, 3, others)

Reviewer comment (verbatim, key points)

Figure 4 visually reports Mann-Whitney $p = 0.016$, Cohen's $d = 0.96$, DeFi mean = 0.043, and DePIN mean = 0.090. However, the caption and revised text report $p = 0.014$, $d = 1.03$, DeFi mean = 0.041, and DePIN mean = 0.091. Figure 5 has title reporting $r = 0.19$, $p = 0.25$, while the caption and abstract report $r = 0.18$, $p = 0.28$. The same goes for other figures, too. My general suggestion is to reproduce all the tables and figures, and make the text reflect the current data.

Response

The reviewer is correct on the specific figures and on the general methodology suggestion. The revised paper adopts a “reproduce all tables and figures from one regression dataset” discipline. All main-text figures have been regenerated from the regression dataset and the visual statistics now match caption and body-text values exactly.

The figures in the revised manuscript are renumbered relative to R1. The R1 figure numbering (3, 4, 5, 6, 7, 8) reflected an earlier iteration that included two theory figures (philosophical-framework architecture; conceptual model) at positions 1 and 2; those theory figures were removed during the empirical reframe and the remaining figures were renumbered 1 through 6 (with a supplementary 2b panel for the three-class sector contrast). The revised figures are:

- **Figure 1** (HHI bar chart across 52 governance tokens; was R1 Figure 3): Holding HHI ranges 0.005 (Hyperliquid) to 0.199 (Livepeer); the DePIN cluster in the upper half is visible.
- **Figure 2** (DePIN versus DeFi sector boxplot; was R1 Figure 4): the figure displays the of-record post-exclusion HHI distribution by sector (DePIN mean 0.067 versus DeFi 0.026, $N = 15 / 15$) and reports the headline sector test under the voter-inclusive staking pass-through treatment (Mann-Whitney $p = 0.028$, Cohen's $d = 0.65$), with the uniform staking-aggregation exclusion noted as the robustness check (Cohen's $d = 0.75$); visual and caption stats match. The figure-versus-caption drift the reviewer identified for this figure is resolved: the revised figure and caption are reconciled to the pass-through headline.

- **Figure 2b** (three-class sector boxplot; new): DePIN versus DeFi versus L1/infrastructure across the $N = 44$ governance-token frame (Kruskal-Wallis), confirming the sector effect generalizes beyond the balanced 15/15 contrast.
- **Figure 3** (insider allocation scatter; was R1 Figure 5): the figure plots the original insider-allocation cohort (Pearson $r = 0.07$, $p = 0.68$, $N = 37$) and the caption notes that the null holds on the current $N = 50$ frame at $r = 0.10$, $p = 0.49$. The figure-versus-caption drift the reviewer identified is resolved; the revised figure and caption both report $r = 0.07 / p = 0.68 / N = 37$ for the cohort scatter. The lead allocation null is the current-frame result ($r = 0.10$, $p = 0.49$, $N = 50$), reported in the abstract and Section 4.4.
- **Figure 4** (delegation-adjusted bar chart; replaces R1 Figure 6): eighteen-protocol delegation amplification chart (thirteen amplify, five disperse) with ratios annotated; visual and caption stats reflect the eighteen-protocol voting-HHI sample.
- **Figure 5** (leave-one-out sector-contrast forest; new): computed under the uniform staking-aggregation exclusion (the robustness-check treatment, Cohen's $d = 0.75$); all 30 leave-one-out iterations remain significant (per-iteration p range 0.006 to 0.031) and the Cohen's d point estimates span 0.68 to 0.92.
- **Figure 6** (subsidy ratio scatter; was R1 Figure 7): Pearson $r = 0.62$ inclusive of Livepeer, $r = 0.07$ excluding Livepeer; $N = 23$; visual and caption stats match.

Reproducibility commitment. Per the reviewer's general suggestion and to prevent recurrence, the revised paper regenerates every body-text statistic, table, and figure from a single regression dataset (shipped as Supplementary File S5) and ships a statistics ledger (Supplementary File S0; new in R2) listing every body-text statistic with its sample size and reporting convention, providing one source of truth for body, tables, figures, captions, abstract, and conclusion.

Manuscript changes: All main-text figures regenerated from the regression dataset; Supplementary File S5 documents the empirical pipeline and figure-regeneration discipline; the regression dataset CSV ships with the replication materials; the statistics ledger ships as Supplementary File S0.

Substantive enhancements beyond Reviewer 1 issues

The revision also includes substantive strengthening that goes beyond the four issues Reviewer 1 raised. These are summarized below for transparency.

Predominant-amplification thesis with structural exceptions. Resolving Issue 1 and extending the voting sample to all protocols with computable proper-weight voting concentration produced the revised delegation finding: of eighteen protocols, thirteen amplify (range 1.6x Drift to 25.6x Polkadot; mean approximately 6.8x, median 4.4x), with five structural exceptions. Relative to R1, the eighteen-protocol table adds Solana-native protocols (Jupiter, Helium, Drift) on a proper-weight basis, the cross-section additions Gnosis, Synthetix, Bittensor, and Polkadot, and the Livepeer orchestrator-bonded-stake governance measure. The five dispersers are ENS at 0.45x (Section 4.5.1), GMX at 0.87x (Section 4.5.1.1), Helium at 0.35x to 0.53x (Section 4.5.4), Jupiter at 0.043x (Section 4.5.4, the most-extreme dispersion outlier), and Livepeer at 0.27x

(Section 4.5, the most pronounced DePIN governance disperser). The Livepeer measure and the strengthened insider layer are detailed in the enhancements below.

Within-Solana governance heterogeneity (Section 4.5.4). Three Solana-native protocols have on-chain voting HHI computable via proper-weight methodology: Jupiter at 0.00546 via per-signer max(weight) aggregation (delegated-vote design); Helium at 0.0261 to 0.0394 (sensitivity range under VSR position-state reconstruction; a 14.8 percent closed-position rate brackets the bounds); Drift at 0.0833 via direct on-chain VoteRecordV2 parsing (VSR lockup-weighted design analogous to Helium). The within-Solana spread (0.005 to 0.083; 15x range) is structurally organized by voting-weight mechanism design as the primary axis: VSR-using protocols (Helium, Drift) cluster at EVM-typical voting HHI; delegated-vote Jupiter is the only Solana protocol exhibiting voting concentration an order of magnitude below the EVM cross-section median. A categorical claim that Solana governance is less concentrated than EVM governance is not supported by the proper-weight Solana sample; the apparent low-concentration finding from vote-count-proxy methodology on Helium was an artifact superseded by lockup-weight reconstruction.

Drift Foundation proposal-authorship versus vote-weight decoupling (Section 4.5.4.1). Drift Protocol exhibits a governance pattern with no close analog in the EVM-protocol cross-section. A single Drift Foundation submitter address (4CL25...wPiiF on Solana) proposed 11 of the 13 DRIFT Improvement Proposals over the analysis window while holding only 0.18 percent of total DRIFT vote weight. Proposal-authorship and vote-weight are structurally decoupled: the Foundation submits proposals, the token-weighted voter pool decides, and the two functions are not co-located in the same address. EVM-protocol top delegates are typically both agenda-setters and vote-weight-holders (Compound Foundation as the Tally top delegate both proposes and votes from a foundation-controlled wallet at 21.5 percent share); the Drift Foundation structure separates these functions explicitly.

Cross-protocol institutional-governance-investor pattern (Section 4.5.5). A candidate fifth concentration axis emerges from cross-protocol aggregation. The Snapshot voter-pool extension aggregated 11,077 unique voter addresses; 1,801 voters (16.3 percent) participate in 2 or more protocols' Snapshot governance. A small institutional-governance-investor class (approximately 10 to 50 addresses) participates in 6 or more protocols' delegate programs simultaneously. The top three institutional cross-protocol delegators collectively exert 17.0 percent combined Snapshot voting power across the L1 DeFi suite: PGov (0x3fb19771...; pgov.eth) at 10.16 percent combined summed share across 7 DAOs; Tane Governance at 4.83 percent across 6 DAOs; Arana Digital at 2.04 percent across 6 DAOs. These are professional governance firms whose business model is providing cross-protocol delegation services to token-holders who do not vote directly; the firms aggregate delegated voting power across multiple protocols simultaneously, creating correlated voting patterns invisible to per-protocol Mann-Whitney or Cohen's d analysis. The Tally-side cross-protocol overlap network is identified as a measurement priority for future work.

Profitability, size, and insider-retention checks (Section 4.6). Secondary checks sharpen the insider-retention interpretation. Larger protocols by fully diluted valuation have fewer insider wallets in their current top-10 holder set (Pearson $r = -0.36$, $p = 0.032$, $N = 36$; Spearman $\rho = -0.31$, $p = 0.063$), and the market-capitalization specification has the same direction at borderline significance (Pearson $r = -0.34$, $p = 0.050$, $N = 33$). Balance-retention specifications are not

significant, indicating a composition-shift interpretation rather than a retained-balance result. Net-deflationary protocols also show no statistically significant difference in insider retention by count or balance (Mann-Whitney $p = 0.57$ and $p = 0.78$ respectively). These checks support the paper's main distinction between launch allocation and current holder composition.

Empirical-paper reframe. The R1 manuscript developed a five-lens philosophical framework integrated with the empirical results; the revised manuscript leads with the measurement contributions and routes the philosophical-interpretation thread to a companion paper (Zukowski 2026e). The reframing preserves every R1 statistical finding. Key surface changes:

- Title: from the R1 submission title “Governance Concentration Beyond Token Allocation: An Institutional Design Analysis of DePIN and DeFi” to “Governance Concentration Beyond Token Allocation: A Protocol-Address-Corrected Cross-Sectional Audit of 52 DePIN and DeFi Protocols”. The lead phrase is preserved exactly for editorial-system continuity; the subtitle is reframed from theory-framing to empirical-audit framing.
- Abstract: leads with the allocation null, sector contrast, subsidy null, delegation amplification, and the PCA methodology contribution; no philosophical-framework framing.
- Section 1.2 (new): one-paragraph bridge to the companion paper Zukowski (2026e) at SSRN registering the Polanyian fourth-fictitious-commodity interpretation; the bridge cites the companion paper rather than developing the argument inline.
- Section 5.6 Contributions: four empirical contributions (PCA methodology; allocation null with insider-retention contrast; corrected sector contrast; predominant delegation amplification with structural exceptions); no two-layer philosophical-framework contribution.
- JEL expansion from seven to nine codes (added C81 Methodology for Collecting Microeconomic Data anchoring the PCA exclusion methodology; added G23 Non-bank Financial Institutions anchoring the DeFi and meta-governance institutional-investor coverage).
- Reference list cleanup: 40 orphan references (residue from the earlier theory-heavy framing) removed from the bibliography; 19 entries migrated to the companion paper's bibliography; Polanyi 1944 preserved as the Tier-1 keeper because Section 1.2 names “Polanyian interpretation” verbatim.
- Section 1.1 motivation paragraph order corrected so setup precedes the findings teaser; “none explains” softened to “none is sufficient as a standalone explanation” because sector membership, insider retention, and delegate-program design do explain parts of the variation.
- Section 2.1 opener rewritten gap-first to lead with the three measurement gaps motivating the paper.
- Section 4.5 paragraph 2 deduplicated to lead with the sector breakdown rather than restating the headline.
- Section 5.2 collapsed to a single design-tractability paragraph naming the Anyone Protocol and ENS counterexamples.
- Figure 1 relocated from Section 3.2 Methods to Section 4.2 Results adjacent to Table 3.

- Section 5.8 split into Section 5.8 Future Research plus a new Section 5.9 Reproducibility and Data Availability.
- Headline-statistic repetition reduced (canonical statements in the abstract, the Section 4.5 lead, and the Section 6 conclusion; abbreviated in the delegation-table and Figure 4 captions).
- Associative-language sweep replaced “predicts”, “predictor”, “predicted by”, “drives”, “driven by”, and “determines” with associative alternatives (“is associated with”, “is consistent with”, “reflects”, “indicator”, “covariate”) outside the Section 5.8 falsifiable-predictions section.
- Section 3.5 Supplemental Instruments stub collapsed; Section 4.6 Additional Concentration Mechanisms stub collapsed with synthesis moved to Section 5.5 Summary.
- Stale section cross-references resolved (Section 4.4 misdirect in the delegation-table footnote; circular self-reference in Section 4.7 robustness; Section 3.9 / Section 4.7 references updated post-renumber).

Falsifiable forward claims plus pre-registration intent (Section 5.8). Five falsifiable predictions operationalized for future panel-data and event-study work: (i) sector persistence (DePIN protocols launching in 2026-2027 will exhibit higher post-distribution holding HHI than DeFi protocols launching in the same period); (ii) delegation amplification universality (any protocol introducing a formal delegate program will produce voting HHI greater than holding HHI within twelve months of launch, with amplification magnitude varying by delegate-program design rather than by sector); (iii) insider-retention persistence (within-protocol panel correlation: protocols whose insider fraction increases will exhibit increased holding HHI); (iv) delegate-program design as moderator (broad-community-delegate distribution design choices yield lower amplification ratios); (v) chain-architecture as trajectory moderator (an age-balanced extended cohort matching Solana and EVM DeFi protocols on distribution-phase maturity discriminates whether chain ecosystem moderates the post-distribution deconcentration trajectory independent of maturity). All five predictions inherit the methodology applied in this paper (PCA-symmetric exclusion via the five-class typology; HHI on post-exclusion top-1,000 holders; Mann-Whitney for two-sample tests; bootstrap 95 percent percentile intervals). Pre-registration commitment: hypothesis specifications, statistical tests, and falsification thresholds will be deposited on the Open Science Framework prior to data collection for any panel-data or event-study extension. Full operationalization detail per prediction in Supplementary File S17.

Longitudinal panel evidence (Supplementary File S7). A companion quarterly HHI panel for 14 governance tokens over 8 quarters post-token-generation event documents that 11 of 14 protocols exhibit monotonic governance HHI decay over the 24-month observation window, with the three exceptions reflecting mechanism-specific lock-in: CRV (vote-escrow), COMP (reservoir-drip emissions), and ENS (claim-window distribution). The longitudinal pattern reinforces the cross-sectional finding that initial allocation is not associated with steady-state concentration: distributions tend to deconcentrate over time across heterogeneous allocation designs.

Data-availability and reproducibility statement (Section 5.9; new). Explicit statement that the 52-protocol governance concentration dataset, the exclusions log (133 protocol-controlled address exclusions across 38 protocols), the regression dataset including Token Terminal subsidy ratios and on-chain financial data, the Hivemapper / io.net / Aethir holder distributions, the

veCRV cumulative deposit data underlying the Section 4.5.2 ve-locking analysis, and the Theil / Atkinson supplementary metrics (Supplementary File S9) are available in the linked replication repository. Replication scripts for each table and figure are provided in Supplementary File S5; the cross-protocol Pearson and Spearman correlations reported across Sections 4.3, 4.4, and 4.6 can be reproduced via the replication scripts using only the linked public-source data.

Insider-classification provenance and the strengthened headline (Sections 3.4, 4.4, 4.6.4).

The insider layer was re-derived from primary on-chain evidence and the full statistical battery re-run against it. The result is reassuring in both directions: the sector headline reproduces and strengthens, and the insider layer that supports it is now source-cited and auditable rather than keyword-coded. First, the headline reproduces from a one-command replication and is robust across six independent insider-classification schemes. The replication package regenerates the powered specification deterministically (one command; no live-API dependency; nine of nine new-cohort holder HHIs reproduce from raw inputs), and the DePIN sector finding holds under every one of six independent insider-classification schemes spanning a keyword-floor lower bound through reliability-gated and adversarially reviewed variants to the evidence-traced classification. The DePIN sector coefficient stays positive and significant under all six, with its two-sided p-value ranging from 0.0058 at the keyword floor to 0.0187 at the most permissive gap-filled variant (0.0139 under the evidence-traced classification); all six fall below 0.02. The insider-retention regressor itself is not significant under any scheme, so the sector channel, not the insider-retention channel, carries the result regardless of how insider holdings are coded (Section 4.6.4). Second, the insider layer is now evidence-traced rather than keyword-coded, and the trace surfaced and corrected a systematic over-count. The classification combines third-party on-chain entity labels (Nansen Address Labels), Blockscout verified-contract labels and Dune exchange tags, vesting-contract bytecode matching, and, for multi-signature treasuries, deployer-and-signer tracing (Section 3.4). The governing rule for multi-signature wallets is team-confirmation: a Safe counts as insider only when its deployer or signing set resolves to the protocol team, founder, or a named investor, rather than treating any multisig as insider by default. This rule corrects an over-count present in keyword-only coding, in which independent large holders' Safes were attributed to the team: of seventeen high-share bare Safes traced across the cross-section, six resolve to protocol teams and eleven to independent holders, with three deployer traces documented as worked exemplars in the replication package. Because the evidence-traced classification corrects this over-count, the sector signal is, if anything, sharpest under the classification (0.0139). The full per-survivor classification spine, the per-scheme robustness table, the seventeen-Safe deployer-and-signer verdicts (six team-confirmed, eleven independent), and the staking-attribution audit (the AAVE and ENA insider-share recovery and the Livepeer orchestrator-level bloc-voting HHI) are documented in Supplementary File S22; the Class 3 versus Class 5 transaction-pattern disambiguation methodology applied in the lower-attribution-coverage cases is in Supplementary File S21.

Multivariate sector coefficient now significant under both the raw and the log measures (Sections 4.4, 4.6.4). The powered specification (Model 4; covariate-complete sample, $N = 50$; HC3 robust standard errors) regresses governance HHI on sector, revenue intensity, and protocol maturity at twelve and a half observations per predictor. Under the log-HHI measure the DePIN coefficient is positive and significant ($p = 0.0197$) and does not attenuate when revenue intensity and maturity are added (both controls individually uninformative, $p = 0.97$ and $p = 0.96$). The strengthening point is that the result no longer depends on the log transform: in untransformed

HHI the DePIN coefficient is the same sign and also significant (coefficient +0.035, $p = 0.030$), so the powered specification holds under both the raw and the skew-correcting log measures. The same-direction contrast on the full unbalanced sample (15 DePIN versus all 24 DeFi) is shown in the sector boxplot and preserves the direction, though its significance is sensitive to the staking-aggregation treatment; it is distinct from the balanced fifteen-versus-fifteen sector headline reported in the main text under the voter-inclusive staking pass-through treatment (Mann-Whitney $p = 0.028$, Cohen's $d = 0.65$), with the uniform staking-aggregation exclusion as the robustness check at Cohen's $d = 0.75$ and the earlier inconsistent-treatment specification (Cohen's $d = 1.05$) reported as inflated rather than as the headline.

Robustness layer added in this revision (Sections 4.3, 4.4, 4.6). A zero-new-data robustness layer, reproducible from committed inputs in the replication package, anticipates the natural follow-up questions about the headline results and answers each on its own terms. On the sector contrast, we now report an honest effect-size triple under three staking treatments rather than a single number: the headline voter-inclusive staking pass-through treatment is a medium effect (Cohen's $d = 0.65$) that is rank-significant (Mann-Whitney $p = 0.028$) and robust across all thirty leave-one-out folds, with a marginal mean-based permutation (p approximately 0.08). We disclose that marginal permutation rather than smoothing it: it is the heavy-tail signature of a single concentrated DeFi-side vote-escrow bloc (Convex-controlled veCRV) that enlarges the mean-difference permutation null, and because the rank-based Mann-Whitney test is insensitive to the magnitude of that tail, we treat the rank-based result as the primary inference. A uniform staking-aggregation exclusion gives a comparable medium effect significant on every test (Cohen's $d = 0.75$, Mann-Whitney $p = 0.018$, label-permutation $p = 0.009$), and the earlier larger effect (Cohen's $d = 1.05$) is reported as inflated by an inconsistent cross-sector staking treatment rather than as the headline, an honest re-characterization rather than a retraction. On the allocation null, a concern that a non-significant correlation could simply be underpowered is addressed directly: the null is statistically equivalent to zero within the envelope the sample is powered to detect (two-one-sided-tests $p = 0.02$ within $|r| = 0.38$), and the launch-design block of team allocation, investor allocation, and protocol maturity is jointly uninformative (omnibus $F(3,41) = 0.66$, $p = 0.58$, explaining 4.6 percent of HHI variance), so the allocation null is a positive bounded-effect result. On the concern that a single cross-section cannot speak to eventual concentration, a descriptive temporal corroboration on the quarterly governance-HHI panel shows launch insider allocation uncorrelated with the 24-month endpoint (Pearson $r = -0.11$, $N = 13$), consistent with the cross-sectional null and reported strictly as descriptive under a no-trajectory-claims scope. On the concern that the sector contrast might hinge on one protocol, influence diagnostics on the powered specification confirm that every one of the leave-one-out refits keeps the DePIN coefficient significant (worst-case $p = 0.022$ in the maturity specification and $p = 0.013$ in the retention specification). And on the insider-retention result, the de-tautology Spearman $\rho = 0.54$ ($N = 34$) carries a bootstrap 95 percent interval of $[0.21, 0.80]$, a permutation $p = 0.001$, all thirty-four leave-one-out folds positive, and a sector-partial rank correlation of 0.47 ($p = 0.005$), so the association is not an arithmetic artifact of insiders being counted in the concentration measure.

Staking-attribution recovers a hidden insider share that the holder snapshot does not register (Section 4.4; the staking-contract exclusion rule is in Section 3.8). Address-level attribution of staked balances back to their underlying stakers makes explicit what the wholesale staking-contract exclusion hides. The stkAAVE staking contract holds 21.67 percent of AAVE

supply, of which roughly 4.4 to 6.3 percent of total supply is insider AAVE (team multi-signature wallets plus the founder) invisible to the top-1000 holder snapshot. The methodologically important point is the direction of the holding-HHI effect: attributing the staked balance back leaves the address-level holding HHI essentially unchanged, and if anything marginally lower, because the recovered insider stake re-enters as several distinct staker addresses rather than as a single block, enlarging the post-exclusion holder denominator more than it concentrates the squared-share sum. The substantive result is therefore the recovered insider share, not a movement in the holding HHI. This extends the paper's argument that the holding measure understates effective control: the concentration the holding HHI does not register surfaces instead in the insider share and in delegated voting power.

Livepeer joins the dispersion exceptions as the most pronounced DePIN governance disperser, with the holding-concentration headline untouched (Section 4.5). Orchestrator-level governance for Livepeer is computed over active orchestrators by total bonded stake (self plus delegated) from the Livepeer protocol subgraph, the apples-to-apples governance-power surface for a bonded-stake-delegation design. The orchestrator bloc-voting HHI is 0.0535 across 100 active orchestrators (as-of Arbitrum block 468186375 / round 4216), 3.7 times lower than Livepeer's raw holding HHI of 0.198868 (ratio 0.27x). Livepeer is therefore the most pronounced DePIN governance disperser, at a 0.27x ratio below Helium's 0.35x to 0.53x range, joining ENS, GMX, Helium, and Jupiter as a structural dispersion exception. This governance-layer dispersion sits on a different axis from the Livepeer subsidy-outlier finding (Section 4.4) and does not disturb the sector headline: Livepeer's holding concentration remains the cross-section maximum (0.199, Table 3), so the DePIN-concentration finding is unaffected by the orchestrator-level voting result. The finding inverts the prior expectation that bonded-stake delegation would concentrate Livepeer governance and is reported as a documented direction-of-effect result rather than a revision to the holding cross-section.

Acknowledgments

We thank Reviewer 1 for the comprehensive cross-surface audit that surfaced the manuscript-versus-data drift in R1 Round 2 and motivated the R2 reproduce-and-sync discipline. The revised manuscript's predominant-amplification finding with structural exceptions (Section 4.5; abstract finding 4) is a direct product of Reviewer 1's framing: applying consistent post-exclusion methodology across Table 4 and the delegation table produced a substantive interpretive change. The broader strengthening of the voting layer (proper-weight Snapshot, Tally, and Solana on-chain governance parsing), the on-chain operating-cost subsidy methodology, and the empirical-paper reframe all extend from the reviewer's general "reproduce all tables and figures and make the text reflect the current data" recommendation. We thank Reviewer 2 for the endorsement of publication.